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AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

October 19, 2016 3:00-5:00 PM

Close Books / Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (40) Consider the motion of a spacecraft subject to the gravity of the Earth modelled as a point mass and its own thrust. Let m be the mass of the spacecraft, $\mu = GM$ be the gravitational parameter of the Earth where G is Newton's gravitational constant and M is the mass of the Earth, $\vec{r}$ be the position vector of the spacecraft, and $\vec{F}$ be the thrust.

   a. (5) Derive the equations of motion of the spacecraft in vector form.

   b. (5) Let $\vec{r} = xi + yj + zk$, $\vec{F} = F_x i + F_y j + F_z k$ and the state vector $\vec{x} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \end{bmatrix}^T$ $= \begin{bmatrix} x & \dot{x} & y & \dot{y} & z & \dot{z} \end{bmatrix}^T$ be defined in the Cartesian coordinate system. Describe the equations of motion in state space form.

   c. (15) Let $\vec{r} = r\vec{e}_r + \cancel{v\vec{e}_v + \lambda\vec{e}_\lambda}$ and $\vec{F} = F_r\vec{e}_r + F_v\vec{e}_v + F_\lambda\vec{e}_\lambda$ be defined in the spherical coordinate system with their components in the order of radial, longitudinal, and latitude direction, and $\vec{\omega}$ be the angular velocity vector of the spacecraft. Describe the equations of motion in the spherical coordinate system.

   d. (5) Show that the following is an equilibrium solution, i.e., an operating point:

$$\begin{cases} r \equiv R > 0 \\ v = \sqrt{\dfrac{\mu}{R^3}}\,t + C, \quad C = \text{constant} \\ \lambda \equiv 0 \end{cases}$$

   e. (10) Linearize the longitudinal (second) and latitude (third) equations of (c) with respect to the operation point given in (d).

2. (20) Consider a simple double integrator system given in the state space form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

   a. (10) If there is no control input, i.e., $u \equiv 0$, discuss the stability of the above system.

   b. (10) Show that the above system is controllable. Then, design your own feedback controller in the form of $u = \alpha x_1 + \beta x_2$ such that the control-augmented system becomes asymptotically stable.

3. (20) A spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the non-dimensionalized equations of motion for the restricted three-body dynamics, answer the following questions:

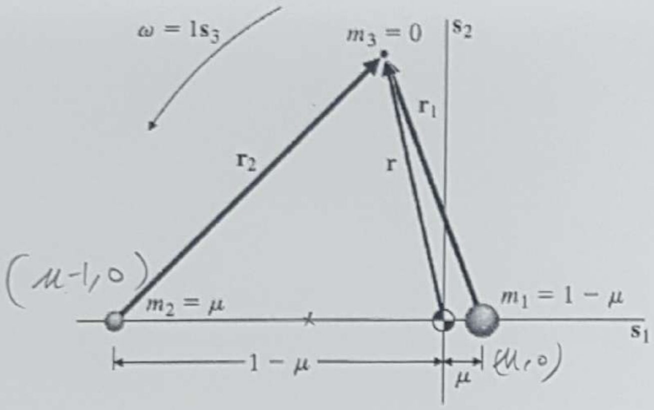

FIGURE 10.1
Geometry of the restricted problem.

$\mu - (\mu - 0.5)$

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$

$$\ddot{y} + 2\dot{x} - y = -\frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3}$$

$$r_1 = \sqrt{(x-\mu)^2 + y^2 + z^2},\ r_2 = \sqrt{(x+1-\mu)^2 + y^2 + z^2}$$

$\mu = 0.0123$ for the Earth-Moon System

a. (15) Is the spacecraft in equilibrium at this midway point between the Earth and Moon? If yes, provide your rationale. If not, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?

b. (5) What is the constant (control) acceleration vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your control acceleration vector in the barycentric coordinate frame.

4. (20) Consider a two-dimensional linear system

$$\Delta\dot{x} = A\Delta x,\ \ \Delta x(0) = \Delta x_0$$

Let $\lambda = \pm j\omega$ and $(\xi, \bar{\xi})$ be the eigenvalues and eigenvectors of the system, respectively. Describe how to find the initial conditions that compose a Halo Orbit with a single frequency of ω.